Response to referee comments: PRD Submission DM13138,

T.Kimpson, S. Suvorova, H. Middleton, C. Liu, A. Melaots, R. Evans, W. Moran

September 3, 2024

Author response to the referee report for the paper DM13138, entitled *'Adaptive cancellation of mains power interference in continuous gravitational wave searches with a hidden Markov model'*.

We thank the referee for their constructive review. We agree with all the comments and have made the recommended changes as described below. The referee comments are reproduced in italics for convenience. We separate the response into "main" comments (indexed alphabetically) and "additional" comments (indexed numerically), following the convention of the referee.

Referee comments

Main comments

A. *In section II, it is not sufficiently clear if the model is intended as a realistic representation of the North American power grid, or just a phenomenological model that is "good enough" for the purpose of subtraction. This should be spelled out better. An authoritative reference on the actual grid's design and/or statistical properties, if the authors can find one, would also be highly valuable for the typical target audience of this article (astrophysicists / data analysts) who are unlikely to have such information at hand.*

Author response:

Yes, we agree with the referee that it is important to specify clearly that the model is intended as a phenomenological model that seeks to reproduce the main qualitative features of the power grid (i.e. an approximately constant central frequency with small amplitude, slow periodic modulations; a voltage with a stochastically wandering phase offset, etc.; see third paragraph in Section IIB). We have followed the referee's suggestion and updated the discussion in Section II to make this point explicitly.

The power grid is controlled by frequency. When load increases frequency drops, more fuel is supplied and frequency rise so demand is met. The statistics are thus the statistical variation of the demand but this is complicated by faults etc. It is a complicated story and there is no simple statistical model for frequency variation. Accordingly, it is difficult to provide a single authoritative reference for the statistical properties of the grid as the referee suggests. We point to two (Folgueras et al. 2017, Schafer et al. 2018) useful papers which discuss thee above issues in the main text.

B. *In the end, it is not clear enough how big the positive impact of the method really is. Especially, and regardless of the answer to point A, it seems necessary to present at least some level of real-data test of the proposed method, beyond the presented simulated noise case. Since LIGO data is readily available via the GWOSC site, this should be straightforward to perform. A clearer summary of the results in abstract and conclusion would then be appreciated.*

Author response:

Yes we agree that testing the method on real data is a valuable addition.

We have updated the main text to also test the method on real data...

TK: this section to be updated out once we have actually done the tests. I will work with Sofia to sort this out.

We also agree with the referee that a clearer summary of the main results quantifying the impact of the method is important to include. We have updated the abstract and the conclusion accordingly. Specifically,

- For the representative system (Table 1) ANC suppresses the 60-Hz interference by approximately 20 dB.
- Without ANC, the HMM tracker is unable to track the wandering CW signal. After ANC, the HMM tracker is able to track the wandering CW signal. The time-averaged RMS error in the HMM tracking after ANC is 0.038Hz for low σ_f and 0.47Hz for high σ_f.
- The detection probability at a 5% false alarm rate, $p_{\rm d}(0.05) \approx 0.5$–$0.6$ across a range of mains power interference parameters and filter parameters.
- TK: +real data results

Additional comments

1. *The abstract currently points out that the validation test is on "synthetic signals", which is perfectly fine, but what is more important is the nature of the noise data.*

   Yes we agree that the behaviour and statistical properties of the noise are important. We have added a qualifying sentence to the abstract that specifies the properties of the noise.

2. *A brief summary of the results would be valuable at the end of the abstract (as per point B above).*

   Author response:
   Yes, we agree that the main results should be better summarised in the abstract. We have updated the abstract accordingly.

3. *A few comments on references the introduction section: The first 3 are very old, describing glitches in initial detectors - consider updating to 2G references like the Davis+ papers already cited later. The standard Advanced Virgo detector reference arxiv:1408.3978 seems to be missing. Please check if [21] (Lee et al) was indeed the intended citation - this seems to be a deep learning waveform model paper, not about artifact identification. At the end of the second paragraph, another important class of line vetoes not cited yet are those based on multi-detector consistency (e.g. doi:10.1088/0031-8949/90/12/125001 and doi:10.1103/PhysRevD.89.064023)*

   Author response:
   We thank the referee for bringing to our attention references which we had previously overlooked, and the erroneous [21] Lee et al. citation. We have now included all the suggested references, removed reference [21], and included discussion on the class of vetos based on multi-detector consistency

4. *In the paragraph below, the statement that candidates are vetoed blindly without checking the strength of the noise contribution is not true for all standard CW pipelines. More nuanced discussions can be found in some recent LVK results papers.*

   Author response:
   Yes we agree with the referee that the previous statement was too broad. We have now updated the paragraph to caveat that some pipelines do have "line robust" algorithms which combine statistics from different detectors to suppress single detector artifacts (e.g. Keitel et al. 2015, Keitel 2016a,b).

5. *Please try using multiple linestyles for better readability of figures 1,6,8,9. In particular, in Fig. 6 it is difficult to discern that the blue x(f) curve actually shows the signal peak, as it is mostly hidden behind the green h(f).*

   Author response:
   We have updated all linestyles in the figures to improve readability, as suggested.

6. *At the end of II.A, there should be an astrophysical reference for the Crab's frequency, not just a software paper. (Or both together, since citing software used is indeed commendable.) Since the Crab's frequency evolves relatively quickly, a reference year would also be appropriate when quoting it with decimal digits.*

   Author response:
   We have updated the reference to the Crab frequency to include both an astrophysical reference and a software reference. We now also quote the Crab frequency for a given reference year, as suggested.

7. *Fig. 1: missing definition of the abbreviation "ASD"*

   Author response:
   The abbreviation ASD is now defined in the caption of Figure 1.

8. *After Eq. (1) the strict separation of frequency and amplitude modulation is misleading, as both parts of the Earth's motion contribute to both modulations.*

   Author response:
   The text has been updated to remove the misleading separation between the two modulations.

9. *Before Eq. (4), when talking about the coupling "in various complicated ways", would this not naturally be expected to cause stochastic overlaps of components with different time delays*

tau? Could the authors comment on whether this is a likely explanation for the observed losses in coherence?

Author response:

The two primary mechanisms by which mains power couples into the strain channel are the shared power supply of the electronic components, and the 60 Hz magnetic field coupling to magnets which are present in certain optical components of the interferometer.

Regardless of the coupling mechanism, the time delay τ is the same; τ is the time delay due to the spatial propagation of the signal between the PEM position and the GW sensing apparatus.

As is discussed in the text, a coherence < 1 occurs due to a non-linear relationship between the signal measurement in the reference and strain channels.

10. *footnote 3: gitlab file URLs can be quite ephemeral. Is there no better (more stable) way to reference this list, such as a GWOSC release with DOI, a LIGO DCC entry, etc?*

    Author response:
    Yes, we agree that is is desirable to provide a more permanent reference to the PEM channels. However, unfortunately at this point in time there is no DOI /DCC reference that covers all nine PEM channels. It is possible to reference the GWOSC O3 auxiliary channel data release, DOI: 10.7935/3bh6-0k98. However this dataset only contains a subset (three) of the (nine) PEM channels. For lack of a better option, we have followed the referee's suggestion and included this DOI as an additional reference.

11. *The statements at the end of II.C and in the Fig. 4 caption contradict each other: are the higher-frequency peaks harmonics of the power mains, or "of different provenance"?*

    Author response:
    Yes, this is a mistake. The higher-frequency peaks are harmonics. We have updated the Fig 4 caption accordingly. We thank the referee for brining this mistake to our attention.

12. *In the coherence discussion, it remains unclear what threshold is considered to be still sufficient, and below which the channel would no longer be considered a reliable witness.*

    Author response:
    Yes, we agree with the referee that the performance of the ANC scheme is directly related to the degree of coherence between the strain and witness channels.

    Specifically, the fraction of possible noise cancellation R for an idealised system is related to the coherence as

    $$R(f) = \frac{1}{1 - |C_{xr}(f)|}$$

    see e.g. DOI: 10.1109/ASPAA.1997.625628 or DOI: 10.3390/s22176591.

    The threshold value of $R(f)$ required in order for HMM algorithms to run successfully is an empirical question that will depend on the properties of the signal.

    We have added a discussion on these points to Section IIC.

13. *In Eq. (12), it is not made explicit what argument the "arg min" is taken over. Would it be appropriate to write $e_k(w)$ with an explicit argument?*

    Author response:
    Yes, we agree with the referee's suggestion. We have updated Equation (12) to specify explicitly that the argmin is taken over the argument $\boldsymbol{w}$ (i.e. the tap weights), which influence the residual $e_k(\boldsymbol{w})$.

14. *In IV.A, can the authors make an argument why neglecting the Earth's movement is not changing qualitatively the signal-vs-line behaviour? Since CW signals are expected to modulate very slowly intrinsically, typically the shape and extent of their detector-frame frequency spectrum should be dominated by these terms.*

    TK: to discuss in person. See also point 18 below.

15. *Also in the same discussion, there is a reference to the F-statistic. This is taking a maximum likelihood over amplitude parameters, not over those related to the Earth's movement. Indeed F-statistic searches typically also include a "demodulation", which is probably what the authors had in mind here. But the way it is currently phrased will confuse a non-expert reader into thinking it maximizes over frequency evolution parameters.*

    Yes, the referee is correct to point out the potential for confusion through the current phrasing. We have rephrased these sentences to make it clear that the F-statistic maximises over h_0, ϕ_0, ι and ψ, leaving a function of the remaining "Doppler parameters".

16. *In IV.B, the definition of signal-to-noise ratio does not match the standard in the CW literature, which usually includes the observation time. The duration is also important for the frequency resolution. Is it the 800s visible in Fig. 7? Please clarify more explicitly.*

    Yes, we agree with the referee that traditionally the definition of SNR in the CW literature does include the value of T_{obs}. In this paper we fix $T_{\text{obs}} = 800\text{s}$ throughout and vary only the ratio h_0/σ_n. Accordingly, we previously felt that for this paper it was sufficient to neglect the influence of T_{obs} in the definition of the SNR. However, we agree with the referee that in order to make better contact with the CW literature it is more appropriate include T_{obs} in the SNR definition; we have updated the relevant definitions in the manuscript accordingly.

    Regarding the question on whether the $T_{\text{obs}} = 800\text{s}$ is visible in Figure 7, one can see that the vertical axis covers the 800s observation period. We have updated the main text to clarify this point more explicitly, as suggested.

17. *The case (ii) "not plotted for brevity" would seem relevant to show - since now fgw lies within the affected frequency range, it sounds like it could be qualitatively different, and more impressive to demonstrate that the method can still deal with it?*

    TK: we can definitely include this as requested, I just need to get the data off of Sofia.

18. *At the end of IV.B, have the authors verified that the Earth Doppler effect is negligible compared to these frequency errors?*

    TK: To discuss in person. Also related to point 14.

19. *In figures 8/9, it would be interesting to also compare with the performance without ANC - is that similar to, or even worse than the "random guess"?*

    It is not possible to run the HMM algorithm without first applying ANC. One can also see this in the left panel of Figure 7, where no dashed white curves are plotted. Consequently, without ANC one cannot obtain a p_{d} or a p_{fa}.

    We have added a sentence to the main text, relating to Figures 8/9, reminding the reader that the HMM can only be deployed in conjunction with the ANC.

20. *Why is the "zero interference ROC curve" shown only in Fig. 9, not 8? Is it understood why the best performing ANC results are still quite far away from this, if before the subtraction looked visually very impressive?*

    Yes, this is a mistake. We agree that the zero interference curve should be plotted for both figures. We have updated Figure 8 accordingly.

    Some of the ANC results are do no perform as well as in the zero interference case, simply because the noise is not being removed perfectly. The performance of the ANC filter can be improved in general by tuning the relevant parameters. For example in the bottom panel of Figure 9, the $\lambda = 1$ ROC curve is very similar to the zero-interference curve, for reasons that are discussed in the text. We have added additional discussion on this point to the main text.

21. *Near the end of V.B, latency issues are discussed. Are these really relevant for CW searches, which are usually run on archival data?*

    The referee is correct to point out that latency is not a problem when dealing with offline searches. We have removed the discussion on latency from the manuscript and now focus exclusively on the computational cost of the method.